Experiment - 5.1.1. Leap Year Checker

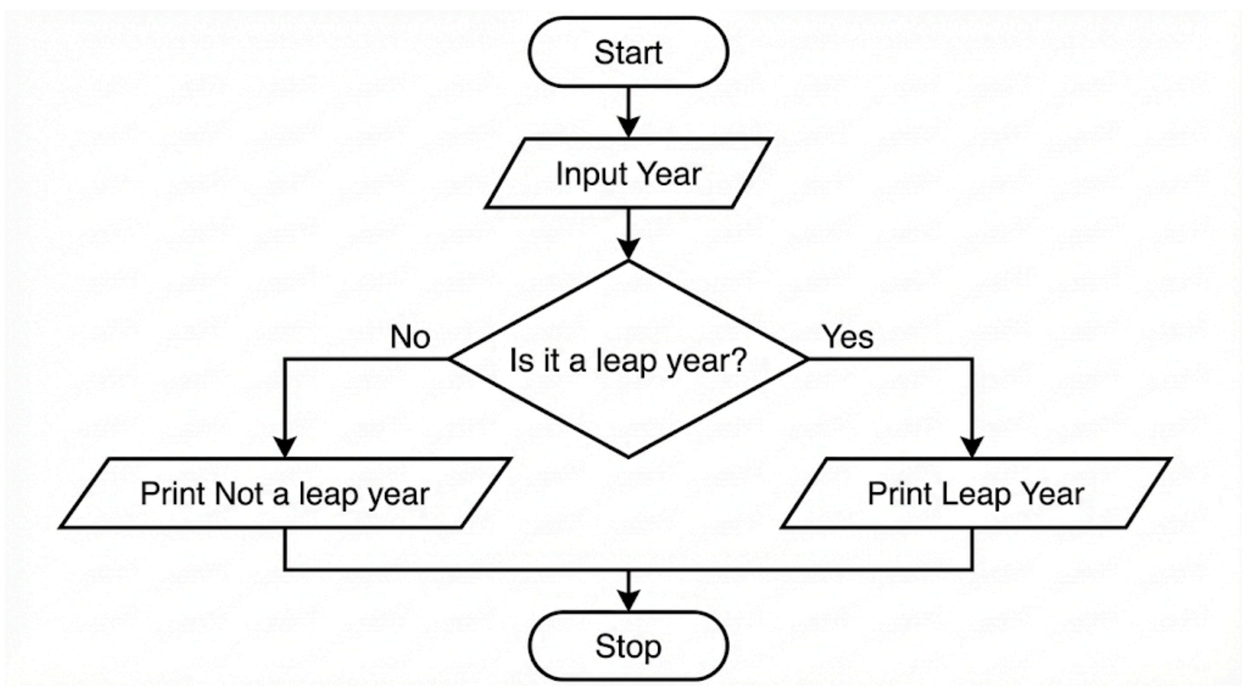
1. Aim

To design and implement a Python program that prompts the user to enter a year and determines whether the given year is a leap year or not, printing the appropriate message.

2. Pseudocode

1. **START**
2. **READ** the input value from the user and convert it to an integer.
3. **STORE** this value in the variable y.
4. **CHECK** if the year y is divisible by 400 (y % 400 == 0).
    - **OR** check if the year y is divisible by 4 AND not divisible by 100 (y % 4 == 0 and y % 100 != 0).
5. **IF** the condition in step 4 is True, **PRINT** "Leap year".
6. **ELSE**, **PRINT** "Not a leap year".
7. **END**

3. Flowchart

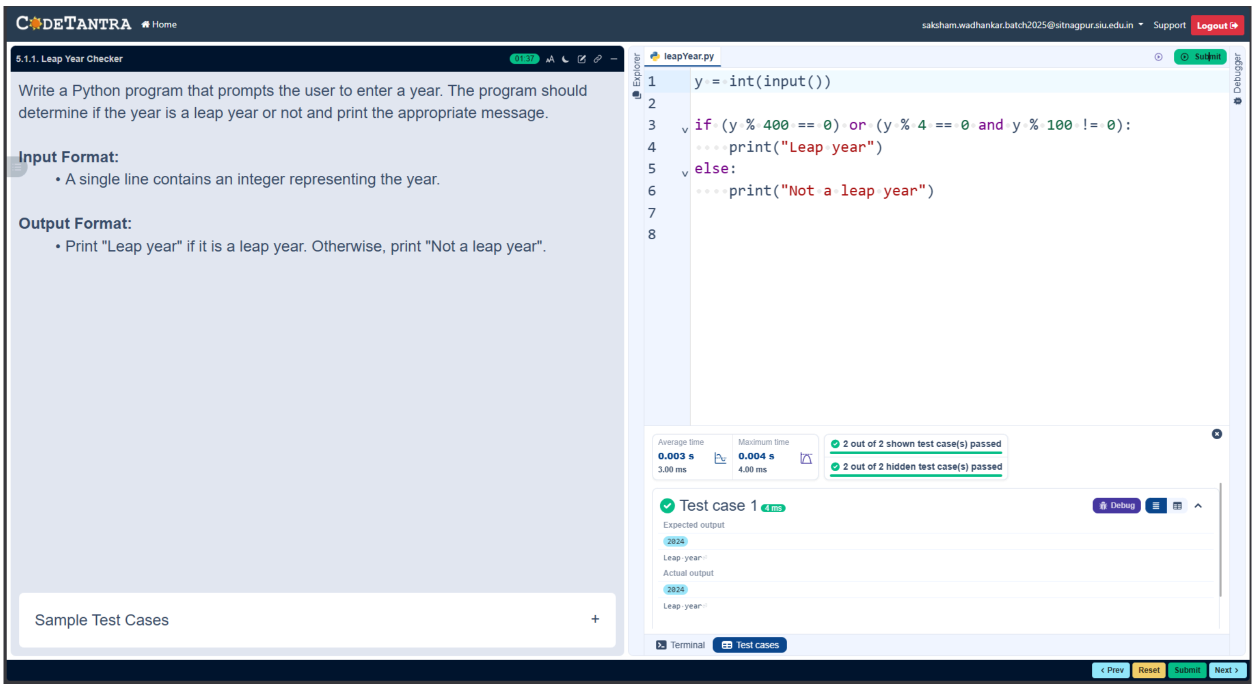


4. Python Program

```
y = int(input())

if (y % 400 == 0) or (y % 4 == 0 and y % 100 != 0):
    print("Leap year")
else:
    print("Not a leap year")
```

5. Experiment Screenshot



Experiment - 5.1.2. Student Grade Based on Aggregate

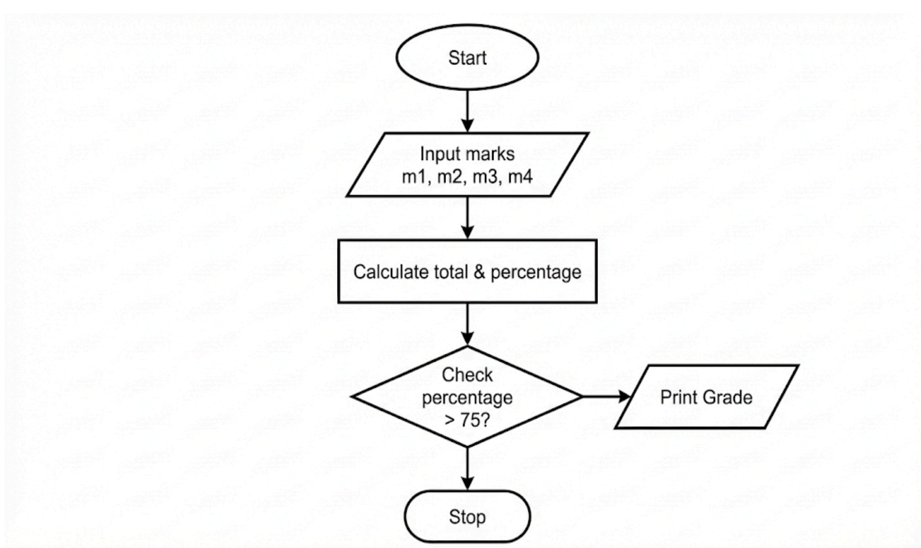
1. Aim

To design and implement a Python program that calculates a student's total marks, aggregate percentage, and final grade based on the marks obtained in four subjects. The program determines the grade using a specific set of percentage-based conditions.

2. Pseudocode

1. **START**
2. **READ** four space-separated integers from the user and **STORE** them in variables m1, m2, m3, and m4.
3. **CALCULATE** the sum of the four subjects and **STORE** it in the variable total (total = m1 + m2 + m3 + m4).
4. **CALCULATE** the aggregate percentage by dividing the total by 4 and **STORE** it in the variable percentage.
5. **CHECK** the value of percentage to determine the grade:
    - **IF** percentage > 75, **STORE** "Distinction" in grade.
    - **ELSE IF** percentage >= 60, **STORE** "First Division" in grade.
    - **ELSE IF** percentage >= 50, **STORE** "Second Division" in grade.
    - **ELSE IF** percentage >= 40, **STORE** "Third Division" in grade.
    - **ELSE**, **STORE** "Fail" in grade.
6. **PRINT** the total marks.
7. **PRINT** the percentage formatted to two decimal places.
8. **PRINT** the grade.
9. **END**

3. Flowchart



4. Python Program

```
# Read four subject marks
m1, m2, m3, m4 = map(int, input().split())

# Calculate total and percentage
total = m1 + m2 + m3 + m4
percentage = total / 4

# Determine the grade
if percentage > 75:
    grade = "Distinction"
elif percentage >= 60:
    grade = "First Division"
elif percentage >= 50:
    grade = "Second Division"
elif percentage >= 40:
    grade = "Third Division"
else:
    grade = "Fail"

# Output the results
print(total)
print(f"{percentage:.2f}")
print(grade)
```

5. Experiment Screenshot